
AIR26 Summer Training Program

Software Prerequisites Guide

Setting up Ubuntu 22.04 LTS for ROS 2 Humble

This document must be completed **before** the ROS sessions begin.

Target Environment

Operating System: Ubuntu 22.04 LTS (Jammy Jellyfish)
ROS Distribution: ROS 2 Humble Hawksbill
Setup Options: VirtualBox **or** Dual Boot Partition

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1 Introduction

Welcome to **AIR26**! Our ROS sessions will be hands-on, so every participant needs a working **Ubuntu 22.04 LTS** environment running **ROS 2 Humble Hawksbill**. ROS 2 Humble is officially supported on Ubuntu 22.04, which is why we standardise on this exact pairing. Using a different Ubuntu version (e.g. 20.04 or 24.04) will lead to package incompatibilities and is not supported during the training.

❗ IMPORTANT: ROS 2 Humble runs **only** on Ubuntu **22.04 LTS**. Installing 20.04, 24.04, or any other release will break package compatibility. This version pairing is a hard requirement, not a recommendation.

This guide walks you through two ways of getting Ubuntu 22.04 onto your machine, followed by the ROS 2 Humble installation that is common to both.

❗ Pick one path and follow it fully:

 **Option 1 – VirtualBox:** Easiest, fully reversible, runs Ubuntu *inside* your current OS. **This is the only option for macOS users.**

 **Option 2 – Dual Boot:** Best performance, Ubuntu runs natively alongside Windows. Recommended if you have a dedicated machine and are comfortable with partitioning.

1.1 Which option should I choose?

Criteria	VirtualBox	Dual Boot
Difficulty	Beginner friendly	Intermediate
Performance	Good (limited GPU)	Excellent (native)
Risk to data	Very low (isolated)	Medium (resizes disk)
macOS support	✓ Yes (see §5)	✗ No
Disk space needed	30–40 GB (virtual disk)	50+ GB (real partition)
Reversibility	Just delete the VM	Requires removing partition

⚠ macOS users: you cannot dual boot Ubuntu, so a virtual machine is your path. See Section 5 for a dedicated walkthrough covering both Intel and Apple Silicon (M1/M2/M3) Macs.

2 Downloads You Need (ISO and Tools)

Download these in advance. The Ubuntu ISO is large (~4–5 GB), so do this on a stable connection well before your session.

2.1 Ubuntu 22.04 LTS Desktop ISO

We use Ubuntu **22.04 LTS** (the latest 22.04.x point release is fine).

Item	Link
Ubuntu 22.04 LTS Desktop (amd64, Intel/AMD)	releases.ubuntu.com/22.04
Ubuntu 22.04 download page	ubuntu.com/download/desktop
Ubuntu 22.04 Server ARM64 (Apple Silicon)	cdimage.ubuntu.com/releases/22.04 (arm64)

i On Intel/AMD machines (Windows, Linux, Intel Macs) download the **Desktop** file named like `ubuntu-22.04.x-desktop-amd64.iso`. **Apple Silicon** note: Ubuntu 22.04 has *no* official Desktop ARM64 image — only the `ubuntu-22.04.x-live-server-arm64.iso` exists. Use that server ARM64 image in UTM (see Section 5); you can add a desktop later if you want a graphical environment.

⚠ **Download only the Ubuntu 22.04 LTS ISO.** Do not install any other version (e.g. 20.04 or 24.04). ROS 2 Humble is officially supported on **22.04 only**, and any other release will cause package incompatibilities during the training.

2.2 Virtualization software (for Option 1 / macOS)

Item	Link
VirtualBox (Windows / Linux / Intel Mac)	virtualbox.org/wiki/Downloads
VirtualBox Extension Pack	virtualbox.org/wiki/Downloads
UTM (Apple Silicon Macs)	mac.getutm.app

2.3 Bootable USB tool (only for Option 2)

Item	Link
Rufus (Windows, recommended)	rufus.ie
balenaEtcher (cross-platform)	etcher.balena.io

⚠ You will also need a **USB flash drive of at least 8 GB** for the dual boot option. Everything on the USB will be erased.

3 Option 1 — Ubuntu 22.04 in VirtualBox

This runs Ubuntu inside a window on your existing OS. Nothing on your host machine is changed or erased.

3.1 Step-by-step setup

1. Install **VirtualBox** and the matching **Extension Pack** from the link above.
2. Open VirtualBox and click **New**.
3. Name it AIR26-Ubuntu, set **Type = Linux, Version = Ubuntu (64-bit)**.
4. Select the Ubuntu 22.04 ISO you downloaded (VirtualBox 7+ can do an unattended install; you may skip that and install manually).
5. **Memory (RAM):** allocate at least **4096 MB** (8192 MB recommended if your machine has 16 GB+).
6. **Processors (CPU):** allocate **2–4 cores**.
7. **Virtual hard disk:** create a new disk, **VDI**, dynamically allocated, size **at least 30–40 GB**.
8. Start the VM. It boots from the ISO; choose **Install Ubuntu**.
9. Follow the installer: language, keyboard, **Normal installation**, and select **Erase disk and install Ubuntu** — this only erases the *virtual* disk, your real files are safe.
10. Create your username and password, finish, and reboot the VM.
11. After first boot, install **Guest Additions** (**Devices** → **Insert Guest Additions CD image**) for full-screen resolution and a shared clipboard, then reboot.

💡 **Tip:** In VM settings under **Display**, enable **3D Acceleration** and raise video memory to 128 MB so RViz and Gazebo render more smoothly.

💡 **Troubleshooting Tip:** If VirtualBox shows *no 64-bit option*, enable virtualization (VT-x/AMD-V) in your BIOS/UEFI. If the VM feels very slow, increase RAM/CPU, enable 3D acceleration, and install Guest Additions.

3.2 Learn it / watch a demo

- <https://youtu.be/Hva8lsV2nTk?si=FCjJr-3cHabLjSSz>
Install Ubuntu 22.04 in VirtualBox – full walkthrough.
- Official docs: [virtualbox.org/manual](https://www.virtualbox.org/manual)
- Ubuntu tutorial: ubuntu.com/tutorials – [Ubuntu on VirtualBox](#)

4 Option 2 — Dual Boot (Ubuntu alongside Windows)

This installs Ubuntu natively on its own partition. You choose between Windows and Ubuntu each time you power on the machine. It gives the best performance but requires care.

⚠️ **CRITICAL:** Dual booting **resizes your disk partitions** — an irreversible operation that can cause data loss if it is interrupted (power cut, forced shutdown). **Back up all important data** to the cloud or an external drive *before* you begin. With a full backup, you have nothing to lose.

4.1 Step-by-step setup

1. **Back up** your files (cloud or external drive).
2. **Free up disk space in Windows:** open *Disk Management* (Win + X → Disk Management), right-click your main partition, choose **Shrink Volume**, and free at least **50 GB** (more is better).
3. **Create a bootable USB:** open Rufus, select your USB drive and the Ubuntu 22.04 ISO, keep the partition scheme as **GPT** for modern UEFI systems, and click **Start**.
4. **Disable Fast Startup** in Windows (Control Panel → Power Options → Choose what the power buttons do) and, if needed, disable **Secure Boot** / enable booting from USB in BIOS/UEFI.
5. **Boot from the USB:** restart and press the boot-menu key (often F12, F10, Esc, or Del — varies by manufacturer), then select your USB drive.
6. Choose **Try or Install Ubuntu**.
7. In the installer, when asked about installation type, select **Install Ubuntu alongside Windows Boot Manager**. This is the safe automatic option that preserves Windows.
8. Allocate the freed space to Ubuntu and continue.
9. Set your timezone, username, and password, then finish and reboot.
10. Remove the USB when prompted. On restart you should see the **GRUB** menu letting you pick Ubuntu or Windows.

💡 **Troubleshooting Tip:** If the **GRUB** menu never appears and the machine boots straight into Windows, disable **Fast Startup** in Windows and check the boot order in your BIOS/UEFI so the Ubuntu boot manager is first.

💡 **Advanced (manual partitioning):** if you prefer **Something else**, a common scheme is: an EFI system partition (if not present), a **root** / partition (ext4, 40+ GB), and an optional **swap** area. Most participants should just use *Install alongside*.

4.2 Learn it / watch a demo

- **Recommended dual boot video:**
 <https://www.youtube.com/watch?v=QKn5U2esuRk>
Step-by-step Ubuntu + Windows dual boot installation.
- Official Ubuntu install tutorial: ubuntu.com/tutorials – [Install Ubuntu Desktop](#)
- Create a bootable USB with Rufus: ubuntu.com/tutorials – [Bootable USB on Windows](#)

5 macOS Users — Running Ubuntu in a Virtual Machine

Macs cannot dual boot Ubuntu, so the supported path on macOS is a **virtual machine (VM)**. The exact steps depend entirely on your Mac's chip, so this section is split into two **fully independent tracks**. Identify your chip first, then follow *only* your track from start to finish — you do not need to read the other one.

How to check your chip: click the  menu → **About This Mac** and read the **Chip** / **Processor** line.

- Reads “**Intel Core i5/i7/...**” → follow **Track A** (Intel-Based Macs).
- Reads “**Apple M1 / M2 / M3**” → follow **Track B** (Apple Silicon Macs).

5.1 Track A: Intel-Based Macs

❗ **IMPORTANT:** On an Intel Mac, download the **amd64** Desktop image only: `ubuntu-22.04.x-desktop-amd64.iso` (link in Section 2). Do *not* download the arm64 image.

Intel Macs use **VirtualBox**. The steps below are complete on their own — you do not need to read any other option in this guide.

5.1.1 Step-by-step setup

1. Download and install **VirtualBox** for **macOS / Intel hosts**, plus the matching **Extension Pack** (links in Section 2).
2. Open VirtualBox and click **New**. Name it **AIR26-Ubuntu**, set **Type = Linux**, **Version = Ubuntu (64-bit)**.
3. Select the `ubuntu-22.04.x-desktop-amd64.iso` you downloaded.
4. **Memory (RAM):** at least **4096 MB** (8192 MB recommended).
5. **Processors (CPU): 2–4 cores**.
6. **Virtual hard disk:** new **VDI**, dynamically allocated, **30–40 GB** minimum.
7. Start the VM; it boots the ISO. Choose **Install Ubuntu**, then **Normal installation** and **Erase disk and install Ubuntu** — this only erases the *virtual* disk, your Mac’s files are safe.
8. Create your username and password, finish, and reboot the VM.
9. Install **Guest Additions** (Devices → Insert Guest Additions CD image) for full-screen resolution and a shared clipboard, then reboot.

💡 **Troubleshooting Tip:** If macOS blocks the VirtualBox installer, open **System Settings** → **Privacy & Security** and click **Allow** for the developer **Oracle**, then run the installer again.

💡 **Troubleshooting Tip:** If the VM is very slow or the display is stuck at a low resolution, increase the allocated RAM/CPU, enable **3D Acceleration** (Display settings), and install **Guest Additions**. If VirtualBox refuses to start a 64-bit VM, update VirtualBox so it matches its Extension Pack version.

Learn it / watch a demo (Intel Macs).

-  <https://youtu.be/Hva8lsV2nTk?si=FCjJr-3cHabLjSSz>
Install Ubuntu 22.04 in VirtualBox – full walkthrough.
- Intel Mac (VirtualBox) tutorial: ubuntu.com/tutorials – **Ubuntu on VirtualBox**

5.2 Track B: Apple Silicon (M1/M2/M3) Macs

❗ **IMPORTANT:** On Apple Silicon, download the **Server arm64** image: `ubuntu-22.04.x-live-server-arm64.iso` (link in Section 2). Ubuntu 22.04 has *no* Desktop arm64 image, so you install the server build and add a desktop afterwards. Do *not* download the amd64 image.

Apple Silicon Macs use **UTM** (free), which runs the native arm64 build of Ubuntu efficiently. VirtualBox does not run x86 Ubuntu well on ARM Macs. The steps below are complete on their own — you do not need to read any other option in this guide.

5.2.1 Step-by-step setup

1. Download **UTM** from mac.getutm.app (or the Mac App Store) and move it to Applications.
2. Download the `ubuntu-22.04.x-live-server-arm64.iso` (link in Section 2).
3. Open UTM → **Create a New Virtual Machine** → **Virtualize** (not Emulate — virtualize is fast on Apple Silicon).
4. Choose **Linux** as the operating system.
5. Under **Boot ISO Image**, browse to and select the Server arm64 ISO.
6. **Memory:** at least **4096 MB** (8192 MB recommended); **CPU Cores:** **4**.
7. **Storage:** **30–40 GB** minimum, then **Save** the VM.
8. Press **Play** (▶). The VM boots the text-based **Server installer**. Use the arrow keys to accept the defaults (use the entire disk — this only affects the *virtual* disk) and create your username and password when prompted.
9. When the install finishes, power off, **remove the ISO** from the VM's drive settings (so it boots from disk), and start again.

💡 **Troubleshooting Tip:** After the install you land at a plain **command-line login** with no graphical desktop — this is normal for a server image, and the next step adds the desktop. If the **ISO will not boot at all**, confirm you downloaded the **arm64** ISO and chose **Virtualize** (not Emulate) in UTM.

Install a graphical desktop (needed for RViz / Gazebo). Log in at the command line and run:

```
# Refresh the package list
sudo apt update

# Install the full Ubuntu desktop + UTM clipboard/display agent
sudo apt install ubuntu-desktop spice-vdagent

# Reboot so the graphical login screen appears
sudo reboot
```

Reboot required: After `sudo reboot` the VM restarts into the normal Ubuntu **desktop login screen**. (If you only ever need ROS on the command line, you may skip `ubuntu-desktop`.)

❗ Apple Silicon runs the **arm64** build of Ubuntu, but ROS 2 Humble installs from the same **apt** repository as on Intel — the commands in Section 6 are identical. Most ROS packages ship arm64 builds, which is exactly why UTM works well here.

Learn it / watch a demo (Apple Silicon).

- <https://www.youtube.com/watch?v=LjLN0OZxvY>
Install Ubuntu on a Mac using a virtual machine – full walkthrough.
- UTM documentation: docs.getutm.app – [Ubuntu guide](#)

❗ **IMPORTANT:** The demo video uses **Ubuntu 24.04**. Follow the same on-screen steps, but install **Ubuntu 22.04 LTS** instead — not 24.04. ROS 2 Humble requires 22.04.

6 Installing ROS 2 Humble (Common to All Options)

Once you are inside Ubuntu 22.04 (VirtualBox, UTM, or dual boot), install ROS 2 Humble. Open a terminal with **Ctrl + Alt + T** and run each block below in order. These steps mirror the official ROS 2 documentation.

❗ **IMPORTANT:** These commands only work on **Ubuntu 22.04**. ROS 2 Humble is not compatible with 20.04 or 24.04. Confirm your version with **lsb_release -a** before continuing.

6.1 1. Set the locale

```
# Ensure UTF-8 locale support is installed
sudo apt update && sudo apt install locales

# Generate and set a UTF-8 locale (ROS 2 expects UTF-8)
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8
```

6.2 2. Enable the required repositories

```
# Add the "universe" repository, which ROS 2 packages depend on
sudo apt install software-properties-common
sudo add-apt-repository universe
```

6.3 3. Add the ROS 2 GPG key and repository

```
# Install curl, then download the official ROS 2 signing key (GPG key)
sudo apt update && sudo apt install curl -y
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key \
  -o /usr/share/keyrings/ros-archive-keyring.gpg
```

```
# Register the ROS 2 apt repository, verified by the key downloaded above
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-
  archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release &&
  echo $UBUNTU_CODENAME) main" \
| sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

6.4 4. Install ROS 2 Humble Desktop

```
# Refresh package sources and upgrade existing packages first
sudo apt update
sudo apt upgrade

# Install the full desktop: RViz, demos, and tutorials
sudo apt install ros-humble-desktop
```

i `ros-humble-desktop` includes RViz, demos, and tutorials. If you want development tools too, also run `sudo apt install ros-dev-tools`.

6.5 5. Source the environment

```
# Auto-load ROS 2 in every new terminal, then load it in the current one
echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

Restart your terminal: The `source` line is applied automatically in **new** terminals. Any terminal you already had open must be closed and reopened (or re-run `source ~/.bashrc`) before `ros2` commands will work in it.

6.6 6. Verify your installation

Open **two** terminals and run a talker/listener demo.

```
# Terminal 1 -- publishes messages
ros2 run demo_nodes_cpp talker
```

```
# Terminal 2 -- should print the messages coming from Terminal 1
ros2 run demo_nodes_py listener
```

If the listener prints the messages published by the talker, **your ROS 2 Humble installation is working correctly. ✓**

💡 Troubleshooting Tip: If you see `ros2: command not found`, the environment is not sourced — run `source ~/.bashrc` or open a fresh terminal (see step 5).

6.7 Official references

- ROS 2 Humble install (Debian packages): [docs.ros.org – Ubuntu Install \(Debs\)](https://docs.ros.org/en/humble/Installation/Ubuntu-Install-Debs.html)
- ROS 2 documentation home: docs.ros.org/en/humble

7 Learning Resources

Get a head start before the sessions with these tutorials.

7.1 Linux / terminal basics

If you are new to Linux, get comfortable with the terminal:

- Ubuntu command line tutorial: [ubuntu.com/tutorials – Command line for beginners](https://ubuntu.com/tutorials/command-line-for-beginners)
-  <https://www.youtube.com/watch?v=oxuRxtrO2Ag>
Linux command line basics for beginners.

8 Pre-Session Checklist

Tick all of these off **before** the first ROS session:

- ☐ I have chosen my setup path (VirtualBox / UTM *or* Dual Boot).
- ☐ Ubuntu **22.04 LTS** is installed and boots successfully.
- ☐ I can open a terminal and connect to the internet from Ubuntu.
- ☐ ROS 2 Humble is installed (`ros-humble-desktop`).
- ☐ `source /opt/ros/humble/setup.bash` is in my `~/.bashrc`.
- ☐ The talker/listener demo works (verification step passed).
- ☐ I backed up important data (dual boot users especially).

9 Getting Help

Most common problems are addressed **inline**, right next to the step they relate to — look for the  **Troubleshooting Tip** callouts throughout this guide (VirtualBox 64-bit/performance, the Oracle privacy prompt, the Apple Silicon command-line login, the dual-boot GRUB menu, and `ros2: command not found`).

 **Still stuck?** Reach out to the AIR26 organising team **before** the session. Include your operating system, your Mac chip (Intel or Apple Silicon) if relevant, and a screenshot of the error so we can help you in time.

See you at AIR26 — happy robotics! 